

What is a Confidence Interval?

Presentation Outline

1 Frequentist Statistics

2 Bayesian Statistics

Confidence Intervals

Parameter value is fixed and unknown.

Confidence Intervals are calculated using Sample Statistics.

- Interval is random - different from sample to sample.
- Over a large number of repeated samples, $(1 - \alpha)\%$ of the intervals will contain the fixed parameter value.

For any given confidence interval, the parameter is either in the interval or not.

The probability $(1 - \alpha)$ describes the process of constructing the confidence intervals.

Credible Intervals

Parameter value is a random variable with a distribution.

Combine prior distribution with data collection to obtain posterior distribution.

Can make probability statements about parameter using posterior distribution.

95% Credible Interval for Parameter: The probability the parameter is in this interval is 0.95.

Connection between Confidence Intervals and Credible Intervals

In almost all cases, you can set up the analysis so that you obtain a credible interval that is exactly the same as the confidence interval.

You can interpret a confidence interval as a credible interval.